

Upgradable Modular Housing



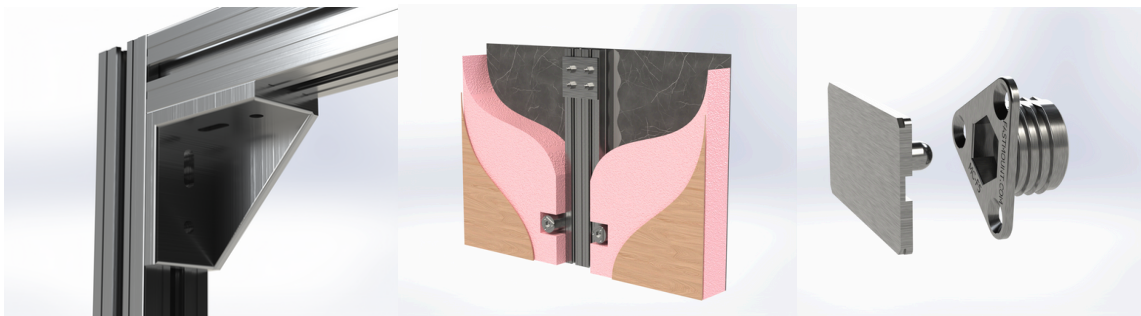
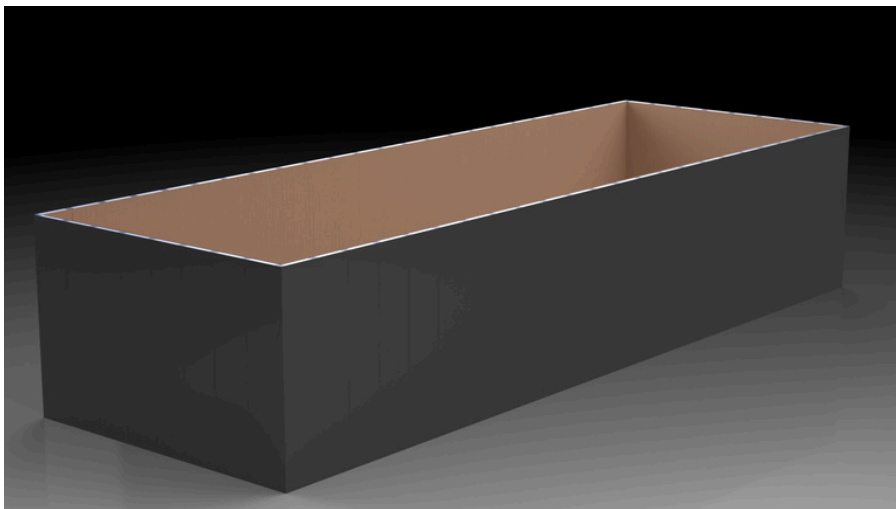
Beauty in every form.



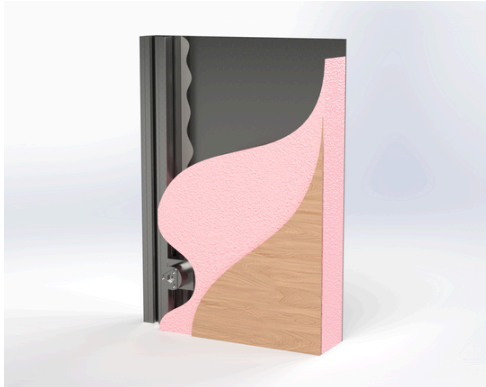
Blake Reed

Context

This project is part of an honours-level industrial design research initiative focused on modular housing innovation. Drawing on extensive industry research, site observations, and compliance analysis, the Proteus™ system was conceptualised as the core architectural framework for the Protea Living™ product line. The design had to address National Construction Code (NCC) Class 1 requirements, factory feasibility, transport and installation logistics, and the growing consumer demand for customisable interiors. The use of T-slot aluminium framing, removable internal panels, and integrated insulation reflects a synthesis of modern methods of construction (MMC) and user-centred design principles.



Challenge



Australia's housing sector is under increasing pressure to provide adaptable, sustainable, and affordable living solutions amid rising material costs, environmental concerns, and demographic shifts.



Traditional construction methods often lack flexibility, are slow to build, and present barriers to future upgrades. The challenge was to develop a wall system that not only meets structural and environmental performance standards but also enables lifecycle adaptability for modular housing.

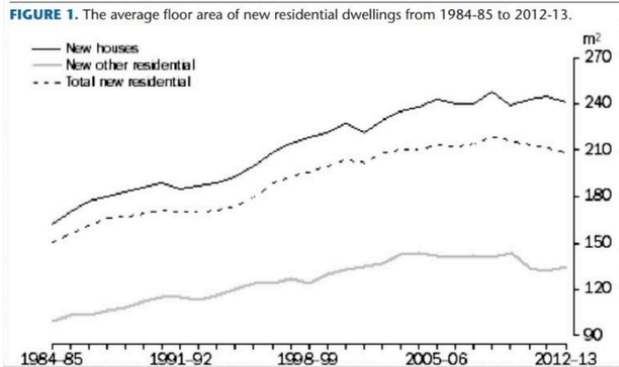
Outcome



The final design solution is a modular, upgradeable wall panel system that enables users to replace, upgrade, or service panels without structural disruption. The Proteus™ system supports diverse cladding options, internal access for utilities, and rapid on-site installation, ensuring durability, weather resistance, and future-proof performance. The system is currently being finalised in CAD for prototyping and fabrication, with all materials selected and compliance considerations met.

Problems

Floor Volume



Wait Times

Average build times for new homes

By property type

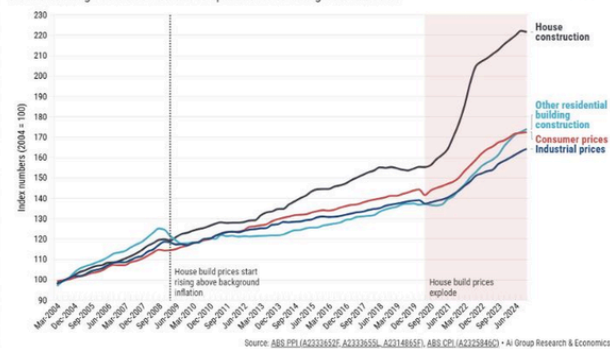
■ Approval to commencement ■ Commencement to completion



Construction Costs

Housing construction costs in Australia

House building costs rise well above apartments or background inflation



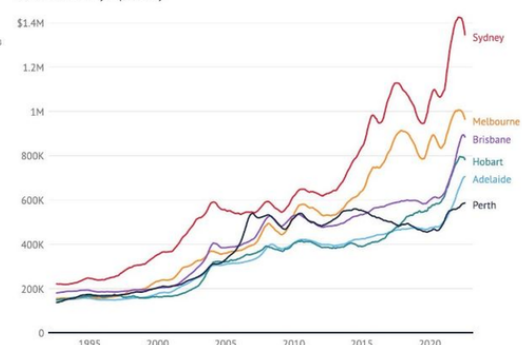
Source: PropTrack analysis of Building Activity, ABS.

PropTrack

House Costs

House price growth over the past three decades

Median values by capital city



Need for innovation

Analysis Of The Current Market

Australia's housing market continues to face intense pressure from rising costs, limited supply, and growing demand. House prices remain near record highs, especially in major cities like Sydney and Melbourne, while construction costs and build times have surged due to material shortages, labour constraints, and inflation. Although average dwelling sizes have plateaued or declined slightly, affordability remains a major barrier for many buyers. In response, there is growing interest in modular and prefabricated construction as faster, more sustainable, and cost-effective alternatives.

Strengths

- Strong population growth and housing demand, especially in urban centres.
- Robust financial and lending systems reduce risk of mortgage default crises.
- Government support programs such as First Home Owner Grants and shared equity schemes.
- Emergence of modular and prefabricated housing, offering cost and time savings.
- Growing sustainability awareness drives demand for energy-efficient and low-emission housing.
- Innovation in construction technologies, including off-site manufacturing, BIM, and smart housing.
- High long-term capital growth potential, particularly in Sydney and Melbourne.

Weaknesses

- Housing affordability crisis, with median prices in Sydney exceeding \$1.3 million.
- Rapid rise in construction costs outpacing inflation and wages.
- Extended build times, particularly for detached houses and apartments.
- Labour shortages and supply chain delays contributing to inefficiencies.
- Low uptake of Modern Methods of Construction (MMC) due to regulatory and cultural barriers.
- Limited medium-density options, with the market skewed toward detached homes or high-rise apartments.
- Planning and zoning constraints, leading to underutilisation of land and slow development.

Current Development Timeline



Define Product Vision And Objectives



Initial sketches and ideas



Material and Manufacture Method Selection



First Prototypes

Currently finalising the design of the Proteus™ modular wall system, which forms the core of the Protea Living™ housing concept. Key milestones include completing the CAD models, selecting aluminium materials, and resolving internal and external wall panel designs with insulation, wiring access, and weather sealing. A custom T-slot bolt and FastMount system allow panels to be removed or reconfigured.

Benefits

Material Input

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras fermentum enim nulla, id sagittis risus accumsan at.

Economic Performance

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras fermentum enim nulla, id sagittis risus accumsan at.

Table 1. Inputs of materials and energy for modular and site-built versions of the home functional unit (only those inputs that differ appreciably between construction methods are shown)

Significantly different construction methods are showing			
	Unit	Construction method	
		Modular	Conventional
Material			
Wood (marriage walls)	lb	3,190	0
Wood wastage	lb	0	3,300
Drywall wastage	lb	1,380-1,600	2,200
Material transport			
Building materials: 16 metric ton truck	miles	110-480	110-1,600
Building materials: 28 metric ton truck	miles	30-290	110-290
Modules to site: 28 metric ton truck	miles	1,200	0
Worker transport			
To factory: car/light duty truck	miles	2,500-13,000	0
To site: car/light duty truck	miles	1,000-3,500	7,800-26,000
Energy use			
Electricity off-site	MWh	2.0-7.2	0
Electricity on-site	MWh	0.1	0.5-1.6
Gasoline (equipment)	MMBTU	24	15-54
Fuel oil (heating off-site)	MMBTU	24	0
Propane (heating off-site)	MMBTU	0.4	0
Natural gas (heating off-site)	MMBTU	0.7	0
Natural gas (heating on-site)	MMBTU	10-18	85-145
Waste management			
Mixed materials	lb	4,580	5,500

Note: lb = pounds; MWh = megawatt hours; MMBTU = million British thermal units (Btu).

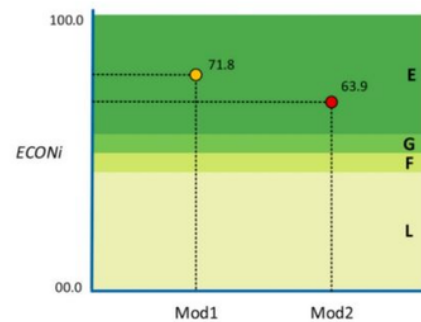


Fig. 8. Performance benchmarking of the case study homes at the economic sustainability dimension level.

Overall Costs

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras fermentum enim nulla, id sagittis risus accumsan at.

Material Waste

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Cras fermentum enim nulla, id sagittis risus accumsan at.

Table 1. Results of quantitative analysis for Project A-SIP and Project B-MOD

Concept	Project A-SIP (\$ CAD/ft ²)	Project B-MOD (\$ CAD/ft ²)
Manufactured elements	\$ 74.60	\$ 75.90
Customization	\$ 4.80	\$ 5.80
On-site interior work	\$ 29.60	\$ 12.40
Other on-site work and utilities	\$ 7.00	\$ 9.50
Total	\$ 116.00	\$ 103.60

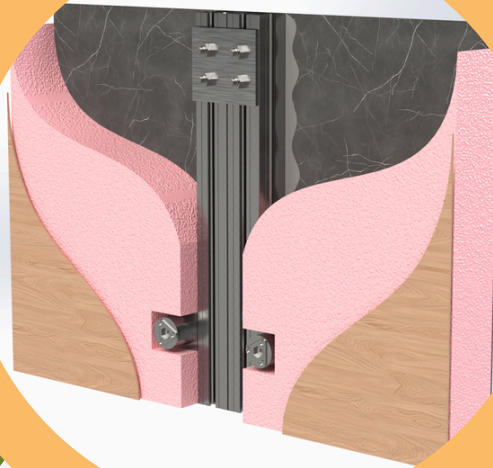
Table 2. Average results from modular and on-site construction, BEES 402 impact assessment method (PAG 2007); rows may not sum due to rounding

	GWP (kg CO ₂ -eq)	Acidification (kg SO ₂ -eq)	Carcinogens (kg 10 ⁻⁶ -eq)	Non-carcinogens (kg 10 ⁻⁶ -eq)	Chronic pollution (kg 10 ⁻⁶ -eq)	Eutrophication (kg N-eq)	Ecotoxicity (kg NCh-eq)	Smog (kg O ₃ -eq)	Water (kg 10 ⁻³ -eq)	Chlorine depletion (kg CFC-11-eq)
Modular average	13,600	5,010	22	47,820	1,290	55	33	83	18,400	229
Material production	603	230	1	3,680	191	4	3	4	1,130	71
Material transport to factory	32	3	<1	14	<1	<1	<1	<1	1	<1
Module transport to site	1,800	596	1	2,960	86	1	1	15	246	1
Factory energy use	6,020	1,820	13	28,200	566	40	15	12	13,000	25
On-site energy use	3,140	1,810	2	5,140	271	2	3	36	900	3
Worker transport to factory	831	193	2	2,990	63	3	4	3	1,020	71
Worker transport to site	1,110	325	3	3,290	107	3	3	6	1,960	108
Waste management	2	1	<1	3	7	<1	<1	<1	1	<1
On-site average	19,500	7,710	30	58,000	2,140	61	47	95	15,500	241
Material production	780	274	2	4,620	234	5	3	4	1,130	29
Material transport to factory	11	4	<1	16	1	<1	<1	<1	2	<1
Module transport to site	0	0	0	0	0	0	0	0	0	0
Factory energy use	0	0	0	0	0	0	0	0	0	0
On-site energy use	11,520	5,470	6	19,450	1,240	12	13	94	4,070	17
Worker transport to factory	0	0	0	0	0	0	0	0	0	0
Worker transport to site	2,180	1,960	22	34,220	689	34	31	32	10,120	695
Waste management	3	2	<1	3	11	<1	<1	<1	1	<1

Note: GWP = greenhouse gases; CO₂-eq = carbon dioxide equivalent units; kg 10⁻⁶ = 1,000 units; 10⁻⁶ = grams; one mole of protein (mol H⁺), DALY = disability-adjusted life years; N = nitrogen; 2,4-D = 2,4-dichlorophenoxyacetic acid; NCh = nitrogen oxides; CFC-11 = trichlorofluoromethane; Chlor kg (kg) = 10⁻³ grams (g) = 10⁻⁶ milligrams (mg) = 2.2046 pounds (lb); one liter (L) = 0.001 cubic meters (m³); 10⁻³ = 0.001 cubic meters (m³).

• Thank You

<https://personal-portfolio-puce-sigma-39.vercel.app/>



blakereed122@gmail.com

+61 450 342 369